



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Department of Computer Science & Engineering

B. Tech. in Computer Science & Engineering

VI Semester

Scheme & Syllabus: 2025-26



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॥ ज्ञानम् सर्वार्थ साधनम् ॥

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Program Name: B. Tech. Computer Science & Engineering

Program Code: CS

Semester VI

Sr. No	Category of Subject	Course Code	Course Name	Teaching Scheme			Evaluation Scheme				Credit	Teaching Mode
				L	T/A	P	CA	MSE	ESE	Total		
1	PCC	CS6T001	Cloud Computing	2	0	0	20	20	60	100	2	Class Board /PPT
2	PCC	CS6T002	Software Testing & Quality Assurance	2	0	0	20	20	60	100	2	Class Board /PPT
3	PCC	CS6T003	AI & Applications	2	0	0	20	20	60	100	2	Class Board /PPT
4	PEC	CS6E003	PEC-III	2	0	0	20	20	60	100	2	Class Board /PPT
5	PEC	CS6E004	PEC-IV	3	0	0	20	20	60	100	3	Class Board /PPT
6	PEC	CS6E005	PEC-V	2	0	0	20	20	60	100	2	Class Board /PPT
7	MDM	CS6M005	LLM Fine Tuning & Optimization	2	0	0	20	20	60	100	2	Class Board /PPT
8	MDM	CS6M006	Multimodelling Development Systems	2	0	0	20	20	60	100	2	Class Board /PPT
9	PCC	CS6L001	Cloud Computing Lab	0	0	2	60	-	40	100	1	PCs
10	PCC	CS6L002	NoSQL Databases Lab	0	0	2	60	-	40	100	1	PCs
11	VSEC	CS6L003	Full Stack Development	0	0	4	60	-	40	100	2	PCs
12	ELC	CS6P003	Field Project / Mini Project-III	0	0	2	60	-	40	100	1	Project
13	JDCAP	CS6T002	JDCAP IV: Skills for Recruitment Success	2	0	0	30	-	20	50	Audit	PCs/ PPT
			Total	18	0	8	430	160	660	1250	22	

Version of the Scheme	JD/ACD/CSE/2023-27/3	Signatures of all BOS members
Year of release of the scheme document:	2025-26	

**Secretary, BOS,
CSE Dept.
JD COEM, Nagpur**

**Chairman, BOS,
CSE Dept.
JD COEM, Nagpur**

**Chairman,
BOS (CSE)
JD COEM, Nagpur**

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6T001	Cloud Computing	2	0	0	2

Prerequisites for the course

1.	Basics of Cloud computing platforms, such as Amazon Web Services, Microsoft Azure, and Google Cloud..
2.	Understand how to host network devices like virtual firewalls, routers, and bandwidth from the cloud

Prior Reading Material/useful links

1.	https://www.mygreatlearning.com/cloud-computing/courses
2.	https://onlinecourses.nptel.ac.in/noc21_cs14/preview

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understanding the Fundamentals of Cloud Computing.
2	CO2	Student will be able to Analysing Cloud Services and Virtualization Technology.
3	CO3	Student will be able to Utilizing Cloud Platforms for Application Deployment.
4	CO4	Student will be able to Managing Cloud Migration and Security.
5	CO5	Student will be able to Exploring Cloud Computing Tools and Future Trends.

Syllabus:

Course Contents		Hours
Unit I	Defining Cloud Computing: Cloud computing in a nutshell, Cloud type - NIST Model, cloud cube model, deployment model, service model, Characteristics of cloud computing, cloud computing stack, open stack	6
Unit II	Understanding Services and Virtualization Technology: Understanding services and applications, defining Infrastructure as a	6

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	Service (IaaS), Platform as a service, Software as a Service, Identity as a Service, Compliance as a Service, Using virtualization technologies, Load balancing and virtualization, understanding Hypervisors, Salesforce.com versus Force.com, SaaS versus PaaS.	
Unit III	Using Cloud Platform: Using Google web services, using Amazon web services, using Microsoft cloud services, Aneka integration of private and public cloud.	6
Unit IV	Cloud Migration: Broad approaches to migration, seven steps model of migration, Applications in the cloud, Application in cloud API, Cloud Security and Storage, Securing the cloud, securing data, working with cloud based storage.	6
Unit V	Cloud Computing Tools and Future Cloud: Open source cloud computing platform - Eucalyptus, Open Nebula, Programming in the cloud Map Reduce Dryad. Future cloud - Future trends in cloud computing, defining the mobile market, using Smart phones with the cloud.	6

Text Books

1.	"Cloud Computing Bible", Barrie Sosinsky; Wiley India Pvt. Ltd
2.	"Cloud Computing - Principals and Paradigms", Rajkumar Buyya, James Broberg, Andrzej Goscinski; Wiley India Pvt. Ltd.
3.	Cloud Computing, A Hands on Approach, Bahga, Madiseti, University Press

Reference Books

1.	"Mastering Cloud Computing", Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Tata McGraw Hill
2.	"Cloud Computing - A practical approach for learning and implementation", A. Shrinivasan, J. Suresh; Pearson
3.	Cloud Computing - Fundamentals, Industry approach and trends", Rishabh Sharma; Wiley India Pvt. Ltd.

Useful Links

1.	https://www.geeksforgeeks.org/cloud-computing/
2.	https://www.techtarget.com/searchcloudcomputing/definition/cloud-computing
3.	https://www.tutorialspoint.com/cloud_computing/cloud_computing_tutorial.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6T002	Software Testing & Quality Assurance	2	0	0	2

Prerequisites for the course

1.	Basic understanding of software engineering principles.
2.	Basic experience with some test automation tools.

Prior Reading Material/useful links

1.	https://www.geeksforgeeks.org/software-testing-vs-quality-assurance/
2.	https://www.tutorialspoint.com/software_testing/software_testing_qa_qc_testing.htm

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understanding the Basics of Software Testing and Its Importance.
2	CO2	Student will be able to gain a thorough understanding of unit testing and Debugging Techniques.
3	CO3	Student will be able to Applying System Integration Testing and Test Design Techniques.
4	CO4	Students will learn to categorize and perform various system tests.
5	CO5	Student will be able to Performing Acceptance Testing and Ensuring Software Quality.

Syllabus:

Course Contents		Hours
Unit I	Basic concepts of Testing: Need of Testing, Basic concepts- errors, faults, defects, failures, objective of testing, central issue in testing, Testing activities, V-Model, Monitoring and Measuring Test Execution, Test tools and Automation, Limitation of Testing.	6
Unit II	Unit Testing: Concepts of Unit Testing, Static Unit Testing, Defect Prevention, Dynamic Unit Testing, Mutation Testing, Debugging, Tools for Unit Testing.	4
Unit III	Control Flow and Data Flow Testing: Control Flow Testing, Control Flow Graphs, Path in Control Flow Graph, Path Selection Criteria,	6

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	Generating Test Input and Data Selection Examples Introduction to Data Flow Testing, Data Flow Graphs, Data Flow Testing Criteria, Comparison of Data Flow Test Selection Criteria.	
Unit IV	System Test Categories and Test Design: Taxonomy of system test, Basic Test, Functionality test, Robustness test, Performance test, Scalability test, Stress test, Load and Stability test, Reliability test, Regression test, Documentation Test. Test Design: Test cases, Necessity of test case documentation, Test case design methods, Functional specification based test case design, Use case bases, Application based test case design, Level of test execution.	7
Unit V	Acceptance Testing and Software Quality: Types of acceptance testing, Acceptance criteria, Acceptance test plan and execution, Special Tests: Client server testing, Web application testing and Mobile application testing, fire view of software quality, ISO-9126 quality characteristics, ISO-9000:2000 software quality standard, ISO - 9000:2000 fundamentals.	6

Text Books

1.	Software Testing and Quality Assurance by Kshirsager Naik and Priyadarshini Tripathi (Wiley)
2.	Software Testing Concepts and Tools by Nageswara Rao Pusuluri (Dream Tech Press).
3.	Software Testing Principles, Techniques and tools, 1st Edition, by M. G. Limaye McGraw Hills.

Reference Books

1.	"Foundations of Software Testing" 2E by Aditya P. Mathur , Pearson Education.
2.	Effective Methods for Software Testing- William E Perry, (Wiley). 2. Software Testing Tools by Dr. K. V. K. K. Prasad (Dream Tech)

Useful Links

1.	https://www.intertek.com/software/quality-assurance/
2.	https://qalified.com/blog/what-is-qa-testing/
3.	https://testsigma.com/blog/software-testing-and-quality-assurance/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6T003	AI & Applications	2	0	0	2

Prerequisites for the course

1.	Knowledge of AI Tools and Frameworks.
2.	Familiarity with big data frameworks.

Prior Reading Material/useful links

1.	https://www.geeksforgeeks.org/applications-of-ai/
2.	https://esdst.eu/arti%E2%81cial-intelligence-the-future-and-its-applications/

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understand the AI and AI Problem.
2	CO2	Student will be able to Analyse the data using predicate logic knowledge.
3	CO3	Student will be able to Solve the problem using Bayes and DST Probabilistic Reasoning
4	CO4	Students will learn to Apply Natural Language Processing kit on given sentence.
5	CO5	Student will be able to Recall and understand the concept of Expert System.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Artificial Intelligence: Definition of AI, History of AI, examples of AI problems. Current trends in Artificial Intelligence, Intelligent Agents, types of agents. Problem solving by search: Uninformed Search: Depth First Search (DFS), Breadth First Search (BFS), Informed Search: Best First Search, A*. Local Search: Hill Climbing, Problem Reduction Search: AO*, Population Based Search: Adversarial Search: Game Playing-Min Max Algorithm, Alpha-Beta Pruning.	7
Unit II	Knowledge Representation: Types of Knowledge, Knowledge	7

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	Representation Techniques, Propositional Logic, syntax, inference, Predicate Logic, Semantic nets, Frames, Knowledge representation issues, Rule based systems.	
Unit III	Reasoning Under Uncertainty: Basics of Probability Theory, Probabilistic Reasoning: Bayes Rules, Probabilistic Reasoning: Bayesian statistic, Dempster-Shafer Theory, Planning: Spare, Block world, Planning with state space search, Representation of Planning, Partial-order Planning.	6
Unit IV	Planning and Learning: Learning: Introduction to Learning, Types of Learning, Rote Learning, Symbol Based Learning, Identification Trees, Explanation Based Learning, Transformational Analogy.	5
Unit V	Applications : Natural Language Processing, Language Models, Text classification, Information Retrieval, information extraction, Expert System: Introduction, Phases in Building Expert Systems.	5

Text Books

1.	Artificial Intelligence: A Modern Approach Stuart R. & Peter Norvig 2nd Edition. Pearson Education
2.	Artificial Intelligence Elaine Rich, Kevin Knight, Shivshankar B Nair, 3rd Edition. McGraw Hill,
3.	Artificial Intelligence Elaine Rich, Kevin Knight, 2nd Edition. Tata McGraw Hill

Reference Books

1.	Artificial Intelligence Patrick H. Winston 3rd Edition Pearson Education.
2.	A First Course in Artificial Intelligence Deepak Khemani McGraw Hill Publication

Useful Links

1.	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
2.	https://www.coursera.org/learn/introduction-to-ai
3.	https://mrcet.com/downloads/digital_notes/IT/(R17A1204)%20Artificial%20Intelligence.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E003A	PEC-III (Middleware Technology)	2	0	0	2

Prerequisites for the course

1.	Basics of Java Programming
2.	Fundamental of client server computing model
3.	Analysis of RMI and JDBC Connectivity.

Prior Reading Material/useful links

1.	https://www.ibm.com/in-en/topics/middleware
2.	https://www.talend.com/resources/what-is-middleware/
3.	https://www.talend.com/resources/what-is-middleware/

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students are able to Choose appropriate client server computing model for given problem.
2	CO2	Students are able to Design a dynamic remote application with RMI and JDBC Connectivity.
3	CO3	Students are able to Develop client server applications using C#.net
4	CO4	Students are able to Select appropriate language for homogeneous and heterogeneous objects.
5	CO5	Students are able to Develop real time projects by combining CORBA and database interfacing

Syllabus:

Course Contents		Hours
Unit I	Introduction to client server computing: Evolution of corporate computing models from centralized to Distributed computing, client server models. Benefits of client server computing , pitfalls of client server Programming. Advanced Java: Review of Java concept like RMI , and JDBC.	8
Unit II	Introducing C# and the .NET Platform: Understanding .NET	8

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	Assemblies, Object –Oriented Programming with C#, Callback Interfaces. Building c# applications: Type Reflection, Late Binding, and Data Access with ADO.NET .	
Unit III	Core CORBA / Java: Two types of Client/ Server invocations-static , dynamic. The static CORBA, first CORBA program, ORBlets with Applets, Dynamic CORBA-The portable count, the dynamic count Existential CORBA: CORBA initialization protocol, CORBA activation services, Introduction to SOA	8
Unit IV	Java Bean Component Model: Events, properties, persistency, Introspection of beans, CORBA Beans. CORBA: Object transaction monitors CORBA OTM's, CORBA OTM's.	3
Unit V	EJB Java Bean: Component Model, EJB Architecture, Session Bean, Java Message Service, Message Driven Bean, Entity Bean	3

Text Books

1.	Client/Server programming with Java and CORBA Robert Orfali and Dan Harkey, John Wiley & Sons , SPD 2nd Edition .
2.	Java programming with CORBA 3rd Edition, G.Brose, A Vogel and K.Duddy, Wiley-dreamtech, India John Wiley and sons.

Reference Books

1.	Distributed Computing, Principles and applications, M.L.Liu, Pearson Education.
2.	Client/Server Survival Guide 3rd edition Robert Orfali Dan Harkey & Jeri Edwards, John Wiley & Sons 3. C# Preciesely Peter Sestoft and Henrik I . Hansen , Prentice Hall of India.

Useful Links

1.	https://books.google.com/books?id=wsnBA4aD2h0C&printsec=frontcover&dq=Middleware+Technologies&hl=en&newbks=1&newbks_redir=1&sa=X&ved=2ahUKEwjeyZ2nmbj-AhXypOkKHUgrDqkQ6AF6BAgHEAI
2.	https://www.google.co.in/books/edition/The Complete Book of Middleware/Gc886KgsdcsC?hl=en&gbpv=1&dq=Middleware+Technologies&printsec=frontcover
3.	https://www.google.co.in/books/edition/Introduction to Middleware/AZgnDwAAQBAJ?hl=en&gbpv=1&dq=Middleware+Technologies&printsec=frontcover
4.	https://www.google.co.in/books/edition/Middleware for Communications/bh8_0FD3qgQC?hl=en&gbpv=1&dq=Middleware+Technologies&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E003B	PEC-III (Soft Computing)	2	0	0	2

Prerequisites for the course

1.	Proficiency with algorithms.
2.	Programming skills in C, C++, or Java, MATLAB, etc.

Prior Reading Material/useful links

1	https://cse.iitkgp.ac.in/~dsamanta/courses/sca/resources/slides/01%20Intro.pdf
2	https://cse.iitkgp.ac.in/~dsamanta/courses/sca/resources/slides/02%20FL%20Introduction.pdf
3	https://cse.iitkgp.ac.in/~dsamanta/courses/sca/resources/slides/06%20ANN%20Introduction.pdf

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to Understanding Soft Computing and its Key Characteristics.
2	CO2	Student will able to Mastering Fuzzy Logic Concepts and Applications.
3	CO3	Student will able to Implementing and Applying Genetic Algorithms.
4	CO4	Student will able to Solving Multi-Objective Optimization Problems Using Evolutionary Algorithms.
5	CO5	Student will able to Exploring Artificial Neural Networks and their Applications.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Soft Computing: Define Soft Computing, Characteristics of Soft computing , Example, Applications of soft computing, Need of soft computing, Elements of soft computing, Soft computing vs hard computing.	3
Unit II	Fuzzy logic: Introduction to Fuzzy logic, Fuzzy sets and membership functions, Operations on Fuzzy sets, Fuzzy relations, rules, propositions, implications and inferences, Defuzzification techniques, Fuzzy logic	8

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	controller design, Some applications of Fuzzy logic.	
Unit III	Genetic Algorithms: Concept of "Genetics" and "Evolution" and its application to probabilistic search techniques, Basic GA framework and different GA architectures, GA operators: Encoding, Crossover, Selection, Mutation, etc, Solving single-objective optimization problems using GAs.	8
Unit IV	Multi-objective Optimization Problem Solving: Concept of multi-objective optimization problems (MOOPs) and issues of solving them. Multi-Objective Evolutionary Algorithm (MOEA), Non-Pareto approaches to solve MOOPs, Pareto-based approaches to solve MOOPs, Some applications with MOEAs.	8
Unit V	Artificial Neural Networks: Biological neurons and its working, Simulation of biological neurons to problem solving, Different ANNs architectures, Training techniques for ANNs, Applications of ANNs to solve some real life problems.	3

Text Books

1.	Soft Computing, D. K. Pratihari, Narosa, 2008.
2.	Neural Networks, Fuzzy Logic and Genetic Algorithms : Synthesis, and Applications, S. Rajasekaran, and G. A. Vijayalakshmi Pai, Prentice Hall of India, 2007.
3.	Neural Networks and Learning Machines, (3 rd Edn.), Simon Haykin, PHI Learning, 2011.
4.	An Introduction to Genetic Algorithms, Melanie Mitchell, MIT Press, 2000.

Reference Books

1.	Fuzzy Logic: A Practical approach, F. Martin, Mc neill, and Ellen Thro, AP Professional, 2000.
2.	Foundations of Neural Networks, Fuzzy Systems, and Knowledge Engineering, Nikola K. Kasabov, MIT Press, 1998.
3.	Practical Genetic Algorithms, Randy L. Haupt and sue Ellen Haupt, John Willey & Sons, 2002.

Useful Links

1.	https://www.geeksforgeeks.org/need-for-soft-computing/
2.	https://www.elprocus.com/soft-computing/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E003C	PEC-III (Digital Image Processing)	2	0	0	2

Prerequisites for the course

1.	Study the fundamental concepts of a digital image processing system.
2.	Categorize various compression techniques.

Prior Reading Material/useful links

1.	https://www.geeksforgeeks.org/digital-image-processing-basics/
2.	https://www.tutorialspoint.com/dip/index.htm
3.	https://www.javatpoint.com/digital-image-processing-tutorial

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to understand the fundamental concepts of a digital image processing system.
2	CO2	Student will able to Analyse images in the frequency domain using various transforms
3	CO3	Student will able to Evaluate the techniques for image enhancement and image restoration.
4	CO4	Student will able to Categorize various compression techniques.
5	CO5	Student will able to Interpret Image compression standards.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Digital Image Processing: Fundamentals of Elements of Digital Image, Image As Data, Pixels, Components Of Digital Image, Types Of Image Representation, Measures Of Image, Neighbors of pixel adjacency connectivity, regions and boundaries, Distance measures, Application Of Digital Image Processing.	8
Unit II	Matlab Basics: Introduction to Data Types, Operators, Matrices, File,	3

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	I/O, Image Processing Toolbox	
Unit III	Image Enhancement Techniques: In spatial domain: Basic gray level transformations, Histogram processing, using arithmetic/Logic operations, smoothing spatial filters, Sharpening spatial filters. In Frequency domain: Introduction to the Fourier transform and frequency domain concepts, smoothing frequency-domain filters, Sharpening frequency domain filters.	8
Unit IV	Image Filtering Techniques: Low Pass Filters – Smoothing, High Pass Filters - Edge Detection, Sharpening; Image Restoration: Noise Models, Model of Image Degradation/Restoration Process, Noise Reduction, Inverse Filtering, M Minimum Mean Square Error (Weiner) Filtering.	8
Unit V	Colour Image processing: Colour fundamentals, Colour models, Representation of Color in Images, Colour transformation, Smoothing and Sharpening, Colour segmentation. Image Morphology: Different Morphological Algorithm, Morphological Measures	3

Text Books

1.	Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing, Pearson Education, Third Edition, 2008.
2.	Anil K. Jain, Fundamentals of Digital Image Processing', Pearson 2002. Gonzalez & Woods - Digital Image Processing Using Matlab
3.	Bhabatosh Chanda and Dwijesh Majumder - Digital Image Processing
4.	Kenneth R. Castleman, Digital Image Processing, Pearson, 2006

Reference Books

1.	Rafael C. Gonzalez, Richard E. Woods, Steven Eddins,' Digital Image Processing using MATLAB', Pearson Education, Inc., 2004.
2.	William K. Pratt, Digital Image Processing' , John Wiley, New York, 2002
3.	D. E. Dudgeon and RM. Mersereau, Multidimensional Digital Signal Processing', Prentice Hall Professional Technical Reference, 1990.

Useful Links

1.	https://www.bharathuniv.ac.in/colleges1/downloads/courseware_ece/course_outcome/core_electives_3/BEC007%20CO%20DIGITAL%20IMAGE%20PROCESSING.pdf
2.	https://www.cis.rit.edu/class/simg712-90/notes/12-Basic_Principles_DIP.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E003D	PEC-III (Deep Learning)	2	0	0	2

Prerequisites for the course

1.	Basic knowledge of how neural networks work.
2.	Skills in cleaning and preparing data for models.

Prior Reading Material/useful links

1.	https://www.geeksforgeeks.org/introduction-deep-learning/
2.	http://neuralnetworksanddeeplearning.com/
3.	https://www.javatpoint.com/deep-learning

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will able to Gain basic knowledge of Neural Networks.
2	CO2	Students will Apply appropriate deep learning algorithms for analysing the data for a variety of problems.
3	CO3	Students will be able to Use the unsupervised deep learning models and analyse the performance.
4	CO4	Students will be able to Gain familiarity with recent trends and applications of Deep Learning.
5	CO5	Students will be Implement deep learning algorithms, understand neural networks and traverse the layers of data abstraction.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Deep Learning: Evolution of AI, Machine Learning vs Deep Learning, Deep Learning types, History of Deep Learning, Model of Artificial Neuron, Activation functions, McCulloch Pitts Neuron, Perceptrons, Forward and Back Propagation, Gradient Descent (GD).	5
Unit II	Deep Learning Models: Introduction to CNNs, CNN operation,s Architectures, Simple Convolution Network, Deep Convolutional Models – LeNet, ResNet, AlexNet and others. Introduction to RNNs, Back	7

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	propagation through time (BPTT), Vanishing and Exploding Gradients, Gated recurrent Unit, Long Short-Term Memory Networks, Dropouts.	
Unit III	Deep Unsupervised Learning: Autoencoders (standard, sparse, denoising, contractive, etc), Motivation for Sampling, Markov Networks, Variational Autoencoders. Introduction of GANs(Generative Modeling) , Different Types of GANs, Components of GANs, Training and Prediction of GANs, Brief on GAN Loss Function, Challenges Faced by GANs, Application of GANs.	7
Unit IV	Regularization for Deep Learning: Overview of Overfitting, Types of biases, Bias Variance Tradeoff Regularization Methods: L1, L2 regularization, Parameter sharing, Dropout, Weight Decay, Batch normalization, Early stopping, Data Augmentation, Adding noise to input and output.	5
Unit V	Applications of Deep Learning: Image segmentation, object detection, Automatic Image Captioning , Capture and Representation, Image Processing, Introduction to NLP, Word Vector representation ,Speech recognition, Video Analytics.	6

Text Books

1.	Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016.
2.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
3.	Deng & Yu, Deep Learning: Methods and Applications, Now Publishers, 2013.

Reference Books

1.	Michael Nielsen, Neural Networks and Deep Learning, Determination Press, 2015.
2.	Neural Networks: A Systematic Introduction, Raúl Rojas, 1996
3.	Pattern Recognition and Machine Learning, Christopher Bishop, 2007

Useful Links

1.	https://deeptai.org/machine-learning-glossary-and-terms/deep-learning
2.	https://journalofbigdata.springeropen.com/articles/10.1186/s40537-021-00444-8

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E004A	PEC-IV (No SQL & New SQL)	3	0	0	3

Prerequisites for the course

1.	Understanding of relational database concepts (tables, keys, constraints).
2.	Ability to write code in any general-purpose language.

Prior Reading Material/useful links

1.	https://www.techtarget.com/whatis/feature/SQL-vs-NoSQL-vs-NewSQL-How-do-they-compare
2.	https://www.xenonstack.com/blog/sql-vs-nosql-vs-newsql
3.	https://www.dremio.com/wiki/newsq/

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to Understand the characteristics and need for NoSQL and NewSQL systems.
2	CO2	Students will be able to model, store, and query data using key-value, document, column-family, and graph databases.
3	CO3	Students will critically evaluate consistency, availability, and partition tolerance in NoSQL and NewSQL systems.
4	CO4	Students will be able to assess performance, fault tolerance, replication, and sharding techniques in distributed databases.
5	CO5	Students will design and implement end-to-end mini-projects or case studies using NoSQL/NewSQL databases for applications in domains.

Syllabus:

Course Contents		Hours
Unit I	Introduction to NoSQL and NewSQL: Need for NoSQL: Limitations of RDBMS, Characteristics of NoSQL Databases (BASE vs ACID), Types of NoSQL Databases: Key-Value, Document, Column-Family, Graph, Overview of NewSQL: Bridging SQL and NoSQL, CAP Theorem and its Implications, Use Cases and Comparison of NoSQL vs NewSQL vs RDBMS	7

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Unit II	Key-Value and Document-Based Databases: Key-Value Stores: Redis, Riak, Document Databases: MongoDB, CouchDB, CRUD Operations in MongoDB, Indexing and Aggregation in Document Stores, Query Languages and Schema Design, Use Cases: Session Storage, User Profiles	7
Unit III	Column-Family and Graph Databases: Column-Family Stores: Apache Cassandra, HBase, Data Modeling and Partitioning in Cassandra, Graph Databases: Neo4j and Graph Theory Basics, Cypher Query Language for Graph Traversal, Applications of Graph Databases: Social Networks, Fraud Detection	7
Unit IV	NewSQL Databases and Distributed Systems: Introduction to NewSQL Systems: Google Spanner, VoltDB, CockroachDB, Architecture and Features of NewSQL, Distributed Database Concepts: Sharding, Replication, Consistency, Query Processing in NewSQL, High Availability and Fault Tolerance in NewSQL	7
Unit V	Performance, Security, and Case Studies: Performance Tuning and Benchmarking NoSQL/NewSQL Systems, Security Aspects: Authentication, Authorization, Data Encryption, Backup, Recovery, and Monitoring Tools, Case Studies: E-commerce (e.g., Amazon using DynamoDB), Real-time Analytics, Financial Applications, Future Trends in NoSQL/NewSQL	8

Text Books

1.	NoSQL Distilled – Pramod J. Sadalage & Martin Fowler
2.	Designing Data-Intensive Applications – Martin Kleppmann
3.	Seven Databases in Seven Weeks – Luc Perkins et al.

Reference Books

1.	Cassandra: The Definitive Guide, Jeff Carpenter, Eben Hewitt, O'Reilly Media
2.	Learning MongoDB, Harsimran Singh, Packt Publishing
3.	Making Sense of NoSQL: A Guide for Managers and the Rest of Us Dan McCreary, Ann Kelly – Manning

Useful Links

1.	https://mrcet.com/downloads/digital_notes/CSE/IV%20Year/SEMANTIC%20WEB&SOCIAL%20NETWORKS.pdf
2.	https://www.google.co.in/books/edition/Semantic_Web_Engineering_in_the_Knowledg/ipN3GWuQXgC?hl=en&gbpv=1&dq=semantic+web+course+objectives&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E004B	PEC-IV (Parallel Computing)	3	0	0	3

Prerequisites for the course

1.	Study Fundamentals of Programming.
2.	Familiarity with algorithm design techniques.

Prior Reading Material/useful links

1.	https://www.tpointtech.com/what-is-parallel-computing
2.	https://onlinecourses.nptel.ac.in/noc24_cs63/preview
3.	https://hpc.llnl.gov/documentation/tutorials/introduction-parallel-computing-tutorial

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will understand the need for performance, types of parallel systems, and differences between concurrent, parallel, and distributed approaches.
2	CO2	Students will be able to write and execute MPI programs involving point-to-point and collective communication, and solve problems.
3	CO3	Students will create thread-based parallel programs, utilize synchronization primitives (mutex, semaphores), and handle issues like thread-safety and false sharing.
4	CO4	Students will utilize OpenMP for loop parallelism, variable scoping, reductions, and task-based parallelism in performance-critical code.
5	CO5	Students will be able to identify and mitigate performance bottlenecks due to cache coherence, false sharing, and load imbalance.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Parallel Computing: Need of Performance, Building Parallel Systems, Why to Write Parallel Programs? How to Write Parallel Programs? Approach: Concurrent, Parallel, Distributed. Parallel Hardware and Parallel Software Background, Modifications to the von Neumann Model, Parallel	7

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	Hardware, Parallel Software, Input and Output, Performance, Parallel Program Design and Writing and Running Parallel Programs	
Unit II	Distributed Memory Programming with MPI: Getting Started, The Trapezoidal Rule in MPI, Dealing with I/O, Collective Communication, MPI Derived Data types, A Parallel Sorting Algorithm.	7
Unit III	Shared Memory Programming with Pthreads: Processes, Threads and Pthreads, Hello World program, Matrix Vector Multiplication, Critical Sections Busy-Waiting, Mutexes, Producer-Consumer Synchronization and Semaphores, Barriers and Condition Variables, Read-Write Locks, Caches, Cache Coherence, and False Sharing and Thread-Safety	7
Unit IV	Shared Memory Programming with OpenMP: Introduction to OpenMP, The Trapezoidal Rule, Scope of Variables, The Reduction Clause, The Parallel For Directive.	7
Unit V	OpenMP, Parallel Program Development and Parallel Algorithms: More About Loops in OpenMP: Sorting, Scheduling Loops, Producers and Consumers, Caches, Cache-Coherence, and False Sharing and Thread-Safety, Two N-Body Solvers, Tree Search.	8

Text Books

1.	An introduction to parallel programming by Peter S Pacheco. 2011. I Edition, Morgan Kaufmann Publishers
2.	Parallel Computers Architecture and Programming, V. Rajaraman, C. Siva Ram Murthy, PHI.
3.	CUDA By Example, Jason Sanders, Edward Kandrot, Addison_Wesley.

Reference Books

1.	Using OpenMP: Portable Shared Memory Parallel Programming, Gabriele Jost and Ruud van der Pas The MIT Press (October 12, 2007)
2.	Using MPI - 2nd Edition: Portable Parallel Programming with the Message Passing Interface, William Gropp and Ewing Lusk, 1999, 2nd edition, MIT Press

Useful Links

1.	https://archive.nptel.ac.in/courses/106/102/106102114/
2.	https://www.nvidia.com/en-us/software/run-ai/#referrer=site&domain=run-ai

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E004C	PEC-IV (Semantic Web)	3	0	0	3

Prerequisites for the course

1.	Study fundamentals of Semantic Web technologies.
2.	Creating structured web documents in XML

Prior Reading Material/useful links

1.	https://www.techtarget.com/searchcio/definition/Semantic-Web
2.	https://www.analyticssteps.com/blogs/what-semantic-web-working-importance-and-applications
3.	https://www.techtarget.com/searchcio/definition/Semantic-Web

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to Understand the fundamentals of Semantic web
2	CO2	Student will able to Creating structured web documents in XML
3	CO3	Student will able to Apply ontology engineering to various problems.
4	CO4	Student will able to Understand Semantic Web query languages (SPARQL)
5	CO5	Student will able to Program semantic applications with Java and Jena API.

Syllabus:

Course Contents		Hours
Unit I	Semantic Web Vision: Today's web, Examples of semantic web from today's web, Semantic web technologies, layered approach Structured web documents in XML: The XML language, Structuring, Namespaces, Querying and Addressing XML documents, Processing	7
Unit II	Describing Web Resources: Introduction, RDF: Basic Ideas, RDF: XML-Based Syntax, RDF serialization, RDF Schema: Basic Ideas, RDF Schema: The Language, RDF and RDF Schema	7

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Unit III	Logic and Inference Rules: Introduction, Monotonic Rules syntax, semantics & examples, Nonmonotonic rules – syntax & examples, Encoding in XML.	7
Unit IV	Ontology Engineering: Introduction, Manual construction of Ontology, Reusing existing ontology, using Semi-automatic methods, Knowledge semantic web architecture	7
Unit V	SPARQL, Ontology Language: SPARQL simple Graph Patterns, Complex Graph Patterns, Group Patterns, Queries with Data Values, Filters OWL Formal Semantics. SchemaWeb Ontology Language: Introduction, OWL language, Examples, OWL in OWL, Future extensions.	8

Text Books

1.	A Semantic web Primer: Grigoris Antoniou and Frank Van Hermelen , MIT Press
2.	Semantic Web programming, John Hebler et.al, Wiley
3.	Foundations of Semantic Web Technologies, Pascal Hitzler, Markus Krötzsch, Sebastian Rudolph, CRC Press
4.	Michael C. Daconta, Leo J. Obrst, and Kevin T. Smith, "The Semantic Web: A Guide to the Future of XML, Web Services, and Knowledge Management", Fourth Edition, Wiley Publishing, 2003.

Reference Books

1.	John Davies, Rudi Studer, and Paul Warren John, "Semantic Web Technologies: Trends and Research in Ontology-based Systems", Wiley and Son's, 2006.
2.	John Davies, Dieter Fensel and Frank Van Harmelen, "Towards the Semantic Web: Ontology- Driven Knowledge Management", John Wiley and Sons, 2003.
3.	Semantic Web and education By Vladan Devedzic · 2006

Useful Links

1.	https://mrcet.com/downloads/digital_notes/CSE/IV%20Year/SEMANTIC%20WEB&SOCIAL%20NETWORKS.pdf
2.	https://www.google.co.in/books/edition/Semantic_Web_Engineering_in_the_Knowledg/ipN3GWuQXgC?hl=en&gbpv=1&dq=semantic+web+course+objectives&printsec=frontcover
3.	https://www.google.co.in/books/edition/Semantic_Web_Services/wglTWruS3UC?hl=en&gbpv=1&dq=semantic+web+course+objectives&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E004D	PEC-IV (Robotic Automation System)	3	0	0	3

Prerequisites for the course

1.	Basic knowledge of circuits, voltage, current, power.
2.	Understanding of stability, sensors, and actuators.

Prior Reading Material/useful links

1.	https://www.techtarget.com/searchcio/definition/RPA
2.	https://www.uipath.com/rpa/robotic-process-automation
3.	https://www.ibm.com/think/topics/rpa

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will Understand the fundamental principles of robotics and industrial automation.
2	CO2	Students will Select and apply appropriate sensors, actuators, and control techniques.
3	CO3	Students will analyse AVR RISC Microcontroller architecture.
4	CO4	Students will Develop and simulate ARM Processor.
5	CO5	Students will be able to Evaluate real-world robotic automation systems and current trends.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Embedded System: Design, Categories of ES, Overview of Embedded System Architecture, Recent Trends in Embedded Systems, Hardware Architecture of Embedded System, Real-time Embedded Systems, Robots and Robotics, Microprocessors and Microcontrollers, Microcontroller or Embedded Controller.	7
Unit II	Robotics: Classification of Robots, Links and Joint, Degree of freedom, Motors-DC motors, Stepper Motors, Servo Motors; Power Transmission-Type of Gears, Robotic Sensors, Applications of Robot, S/w used for Robot	7

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	programming.	
Unit III	The AVR RISC microcontroller architecture: Introduction, AVR family architecture, register file, Pin diagram of AVR, memory organization, I/O ports, timers, USART, Interrupt structure.	7
Unit IV	ARM Processor: Fundamentals, Registers, current program status register, pipeline concept, Interrupt and the vector table	7
Unit V	AI IN ROBOTICS: Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics.	8

Text Books

1.	Subrata Ghoshal, "Embedded Systems & Robots", Cengage Learning
2.	Stuart Russell, Peter Norvig, "Artificial Intelligence: A modern approach", Pearson Education, India 2003.
3.	ARM System Developer's Guide: Designing and Optimizing System Software- Andrew N. Sloss, Dominic Symes, Chris Wright, Elsevier Inc., 2007

Reference Books

1.	Robot Operating System (ROS) for Absolute Beginners – Lentin Joseph, Apress.
2.	Programmable Logic Controllers – Frank D. Petruzella, McGraw Hill.
3.	Modern Robotics: Mechanics, Planning, and Control – Kevin M. Lynch & Frank C. Park.

Useful Links

1.	https://robotnik.eu/what-is-the-difference-between-automation-and-robotics/
2.	https://sitams.org/wp-content/uploads/2023/05/11.18MEC325-Automation-and-Robotics-Lecture-notes.pdf
3.	https://www.techtarget.com/searchdatacenter/definition/parallel-processing

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E005A	PEC-V (Data Warehousing & Mining)	2	0	0	2

Prerequisites for the course

1.	Understand the basics of Data warehousing and Data Mining.
2.	Study the tools and techniques required in Data warehousing and Data Mining.

Prior Reading Material/useful links

1	https://www.topcoder.com/thrive/articles/data-warehousing-and-data-mining
2	https://www.vssut.ac.in/lecture_notes/lecture1428550844.pdf

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will able to Understand the fundamental concepts, architecture, and components of data warehousing and business analysis.
2	CO2	Students will be able to Apply various data preprocessing and transformation techniques and understand the architecture and functionalities of data mining systems.
3	CO3	Students will be able to Analyse and implement classification and prediction techniques.
4	CO4	Students will be able to Evaluate and compare different clustering and outlier detection methods.
5	CO5	Students will be able to Explore and apply data mining techniques to complex data types

Syllabus:

Course Contents		Hours
Unit I	Data Warehousing and Business Analysis: Data warehousing Components, Building a Data warehouse, Data Warehouse Architecture, DBMS Schemas for Decision Support, Data Extraction, Cleanup, and Transformation Tools, Metadata, reporting, Query tools and Applications, Online Analytical Processing (OLAP), OLAP and Multidimensional Data Analysis.	6
Unit II	Data Mining: Data Mining Functionalities, Data Preprocessing, Data Cleaning Data Integration and Transformation, Data Reduction, Data Discretization and	6

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	Concept Hierarchy Generation, Architecture of A Typical Data Mining Systems, Classification Of Data Mining Systems. Association Rule Mining: Efficient and Scalable Frequent Item set Mining Methods, Mining Various Kinds of Association Rules, Association Mining to Correlation Analysis, Constraint-Based Association Mining.	
Unit III	Classification and Prediction: Issues Regarding Classification and Prediction Classification by Decision Tree Introduction, Bayesian Classification, Rule Based Classification, Classification by Back propagation, Support Vector Machines, Associative Classification, Lazy Learners, Other Classification Methods, Prediction, Accuracy and Error Measures, Evaluating the Accuracy of a Classifier or Predictor, Ensemble Methods, Model Section.	6
Unit IV	Cluster Analysis: - Types of Data in Cluster Analysis, A Categorization of Major Clustering Methods, Partitioning Methods, Hierarchical methods, Density-Based Methods, Grid-Based Methods, Model-Based Clustering Methods, Clustering High-Dimensional Data, Constraint-Based Cluster Analysis, Outlier Analysis.	6
Unit V	Mining Object, Spatial, Multimedia, Text and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects, Spatial Data Mining, Multimedia Data Mining, Text Mining, Mining the World Wide Web.	4

Text Books

1.	Jiawei Han, Micheline Kamber and Jian Pei "Data Mining Concepts and Techniques", Third Edition, Elsevier, 2011.
2.	Pang-Ning Tan, Michael Steinbach and Vipin Kumar "Introduction to Data Mining", Pearson Education, 2007.
3.	M.H. Dunham, "Data Mining Introductory and Advanced Topics", Pearson Education.

Reference Books

1.	Alex Berson and Stephen J. Smith "Data Warehousing, Data Mining & OLAP", Tata McGraw – Hill Edition, Tenth Reprint 2007.
2.	K.P. Soman, Shyam Diwakar and V. Ajay "Insight into Data mining Theory and Practice", Eastern Economy Edition, Prentice Hall of India, 2006.
3.	G. K. Gupta "Introduction to Data Mining with Case Studies", Eastern Economy Edition, Prentice Hall of India, 2006.

Useful Links

1.	https://bmsce.ac.in/Content/IS/Big_Data_Analytics_-_Unit_1.pdf
2.	https://www.solvexia.com/blog/big-data-analysis-methods

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E005B	PEC-V (Grid Computing)	2	0	0	2

Prerequisites for the course

1.	Basic understanding of networking concepts.
2.	Knowledge of client-server architecture and network communication.

Prior Reading Material/useful links

1.	https://www.spiceworks.com/tech/cloud/articles/what-is-grid-computing/
2.	https://www.tutorialspoint.com/grid-computing
3.	https://aws.amazon.com/what-is/grid-computing/

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to Understand the fundamental concepts and architecture of Grid Computing.
2	CO2	Student will able to Analyze various grid monitoring architectures and tools.
3	CO3	Student will able to Explain and apply security mechanisms in grid environments.
4	CO4	Student will able to Evaluate grid scheduling and resource management techniques.
5	CO5	Student will able to Explore grid middleware tools and assess their architectures.

Syllabus:

Course Contents		Hours
Unit I	Concepts and Architecture: Introduction-Parallel and Distributed Computing-Cluster Computing-Grid Computing-Anatomy and Physiology of Grid-Review of Web Services-OGSA-WSRF.	6
Unit II	Grid Monitoring: Grid Monitoring Architecture (GMA) - An Overview of Grid Monitoring Systems-GridICE – JAMM - MDS-Network Weather Service-R-GMA-Other Monitoring Systems-Ganglia and GridMon.	6

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Unit III	Grid Security and Resource Management: Grid Security-A Brief Security Primer-PKI-X509 Certificates-Grid Security-Grid Scheduling and Resource Management-Scheduling Paradigms-Working principles of Scheduling -A Review of Condor,SGE, PBS and LSF-Grid Scheduling with QoS.	6
Unit IV	Data Management and Grid Portals: Data Management-Categories and Origins of Structured Data-Data Management Challenges-Architectural Approaches-Collective, Data Management ServicesFederation Services-Grid Portals-First-Generation Grid Portals-Second-Generation Grid Portals.	6
Unit V	Grid Middleware: List of globally available Middlewares - Case Studies-Recent version of Globus Toolkit and gLite - Architecture, Components and Features.	6

Text Books

1.	Frederic Magoules and Jie Pan, "Introduction to Grid Computing" CRC Press, 2009.
2.	JoshyJoseph, CraigFellenstein—Grid Computing, Pearson Education, 2004.
3.	Fran Berman, Geoffrey C. Fox, Anthony J.G Hey Grid Computing making the global infrastructure a Reality, Wiley

Reference Books

1.	Ian Foster, Carl Kesselman, "The Grid: Blueprint for a New Computing Infrastructure", 2nd Edition, Morgan Kaufmann
2.	Daniel Minoli, "A Networking Approach to Grid Computing", John Wiley Publication, 2005
3.	Barry Wilkinson, "Grid Computing: Techniques and Applications", Chapman and Hall, CRC, Taylor and Francis Group, 2010

Useful Links

1.	https://www.kngac.ac.in/elearning-portal/ec/admin/contents/4_18KP2CSELCS2:A_2021012803115174.pdf
2.	https://aits-tpt.edu.in/wp-content/uploads/2022/06/GCC-min.pdf
3.	https://indiafreenotes.com/grid-computing-concepts-architecture-applications-challenges-future/
4.	https://www.scribd.com/document/323686120/Grid-and-Cloud-Computing-Notes

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E005C	PEC-V (Augmented Reality)	2	0	0	2

Prerequisites for the course

1.	Basic knowledge & Information of virtual reality.
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Prior Reading Material/useful links

1.	https://www.techtarget.com/whatis/definition/augmented-reality-AR
2.	https://arvr.google.com/ar/
3.	https://www.ptc.com/en/technologies/augmented-reality

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to understand the basic concept and framework of virtual reality
2	CO2	Student will able to categories the technology for multimodal user interaction and perception in VR.
3	CO3	Student will able to evaluate algorithmic strategies to solve a given problem.
4	CO4	Student will able to apply VR Tools in real time environment.
5	CO5	Student will able to implement application of AR/VR technology with hands on experience through more informative and practical exploration.

Syllabus:

Course Contents		Hours
Unit I	Introduction: VR and AR Fundamentals, Differences between AR/VR Selection of technology AR or VR AR/VR characteristics Hardware and Software for AR/VR introduction. Requirements for VR/AR. Benefits and Applications of AR/VR. AR and VR case study.	6
Unit II	Fundamentals of Computer Graphics: Real time rendering technology; Principles of Stereoscopic Display; Software and Hardware Technology on Stereoscopic Display.	6
Unit III	Software Technologies: Database, World Space, World Coordinate,	8

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	World Environment, Objects, Geometry, Position / Orientation, Hierarchy, Bounding Volume, Scripts and other attributes, VR Environment, VR Database, Tessellated Data, LODs, Cullers and Occluders, Lights and Cameras, Scripts, Interaction, Simple, Feedback, Graphical User Interface, Control Panel, 2D Controls, Hardware Controls, Room / Stage / Area Descriptions, World Authoring and Playback, VR toolkits, Available software in the market (Unity and Vuforia based) - Case Studies in AR, VR – Industrial applications, medial AR/VR, education and AR/VR.	
Unit IV	Geometric Modeling; Behavior Simulation; Physically Based Simulation Haptic & Force Interaction in Virtual Reality Concept of haptic interaction: Principles of touch feedback and force feedback; Typical structure and principles of touch/force feedback facilities in applications	6
Unit V	VR Development Tools: Frameworks of Software Development Tools in VR; Modeling Tools for VR; X3D Standard; Vega, Multi Gen	4

Text Books

1.	Burdea, G. C. and P. Coffet. Virtual Reality Technology, Second Edition. Wiley-IEEE Press, 2003/2006.
2.	Alan B Craig, William R Sherman and Jeffrey D Will, Developing Virtual Reality Applications: Foundations of Effective Design, Morgan Kaufmann, 2009.
3.	Gerard Jounghyun Kim, Designing Virtual Systems: The Structured Approach, 2005.

Reference Books

1.	Doug A Bowman, Ernest Kuijff, Joseph J LaViola, Jr and Ivan Poupyrev, 3D User Interfaces, Theory and Practice, Addison Wesley, USA, 2005.
2.	Oliver Bimber and Ramesh Raskar, Spatial Augmented Reality: Merging Real and Virtual Worlds, 2005.

Useful Links

1.	https://mobidev.biz/blog/augmented-reality-development-guide
2.	https://firsthand.co/professions/augmented-reality-developers/requirements
3.	https://www.coursera.org/learn/augmented-reality

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6E005D	PEC-V (Compiler Design)	2	0	0	2

Prerequisites for the course

1.	Familiarity with algorithm analysis.
2.	Understanding of how code translates into machine instructions.

Prior Reading Material/useful links

1.	https://www.tutorialspoint.com/compiler_design/compiler_design_quick_guide.htm
2.	https://www.tpointtech.com/compiler-tutorial
3.	https://www.tutorialspoint.com/compiler_design/index.htm

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will able to Understand the different phases of a compiler and their roles in the compilation process.
2	CO2	Student will able to Apply top-down and bottom-up parsing techniques to analyze programming language syntax.
3	CO3	Student will able to Design and implement lexical analyzers using regular expressions and tools like LEX.
4	CO4	Student will able to Generate intermediate code representations and perform syntax-directed translation.
5	CO5	Student will able to Analyze and apply code optimization and data flow analysis techniques.

Syllabus:

Course Contents		Hours
Unit I	Overview of Compilation: Phases of Compilation – Lexical Analysis, Regular Grammar and regular expression for common programming language features, pass and Phases of translation, interpretation, bootstrapping, data structures in compilation – LEX lexical analyzer generator.	5
Unit II	Top down Parsing: Context free grammars, Top down parsing – Backtracking, LL (1), recursive descent parsing, Predictive parsing,	7

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	Pre-processing steps required for predictive parsing. Bottom up parsing: Shift Reduce parsing, SLR, CLR and LALR parsing, Error recovery in parsing, handling ambiguous grammar, YACC –automatic parser generator.	
Unit III	Semantic analysis: Intermediate forms of source Programs – abstract syntax tree, polish notation and three address codes. Attributed grammars, Syntax directed translation, Conversion of popular Programming languages language Constructs into Intermediate code forms, Type checker.	6
Unit IV	Symbol Tables: Symbol table format, organization for block structures languages, hashing, tree structures representation of scope information. Block structures and non block structure storage allocation: static, Runtime stack and heap storage allocation, storage allocation for arrays, strings and records. Code optimization: Consideration for Optimization, Scope of Optimization, local optimization, loop optimization, frequency reduction, folding, DAG representation.	7
Unit V	Data flow analysis: Flow graph, data flow equation, global optimization, redundant sub expression elimination, Induction variable elements, Live variable analysis, Copy propagation. Object code generation: Object code forms, machine dependent code optimization, register allocation and assignment generic code generation algorithms, DAG for register allocation.	5

Text Books

1.	Principles of compiler design -A.V. Aho . J.D.Ullman; Pearson Education.
2.	Modern Compiler Implementation in C- Andrew N. Appel, Cambridge University Press.

Reference Books

1.	Engineering a Compiler-Cooper & Linda, Elsevier.
2.	Compiler Construction,- LOUDEN, Thomson.

Useful Links

1	https://www.tpointtech.com/compiler-introduction
2	https://gcekbpatna.ac.in/assets/documents/lecturenotes/lectures_note_on_compiler_design.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6M005	MDM-V (LLM Fine Tuning & Optimization)	2	0	0	2

Prerequisites for the course

1.	Basic knowledge of Large Language Models.
2.	Awareness of the Neural Network & Deep Learning.

Prior Reading Material/useful links

1	https://arxiv.org/abs/2104.10350
2	https://huggingface.co/docs/transformers
3	https://huggingface.co/docs/transformers

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able to Understand the architecture and working principles of Large Language Models.
2	CO2	Students will be able to Apply different fine-tuning methods to adapt pre-trained LLMs for specific tasks.
3	CO3	Students will be able to Implement parameter-efficient techniques like Adapters and LoRA to optimize resource use.
4	CO4	Students will be able to Utilize optimization methods such as quantization and distillation for efficient deployment.
5	CO5	Students will be able to Use modern frameworks and tools to train, fine-tune, and evaluate LLMs effectively.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Large Language Models: Overview of Generative AI and Foundation Models, Architecture of LLMs: Encoder, Decoder, and Encoder-Decoder, Pre-training and Fine-tuning concepts Challenges in LLM training (data, compute, bias, and efficiency)	6

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Unit II	Fundamentals of Fine-Tuning: Transfer Learning and Domain Adaptation, Full Fine-tuning vs. Feature-based Fine-tuning, Dataset preparation and preprocessing for fine-tuning, Evaluation metrics and validation techniques	6
Unit III	Parameter Efficient Fine-Tuning (PEFT): Concept and Need for PEFT, Methods: Adapters, Prefix Tuning, Prompt Tuning, Low-Rank Adaptation (LoRA) and its applications, Combining PEFT methods for efficient model customization	6
Unit IV	Optimization Techniques: Optimizers: Adam, AdamW, Adafactor Gradient clipping, learning rate scheduling, and mixed precision training, Knowledge Distillation and Quantization, Pruning and Compression for deployment efficiency	6
Unit V	Advanced Topics and Tools: Distributed Training: Data, Model, and Pipeline Parallelism, Efficient Memory Management: Flash Attention, Evaluation and Monitoring of Fine-tuned LLMs, Tools and Frameworks: Hugging Face Transformers, PEFT, DeepSpeed, PyTorch Lightning	6

Text Books

1. Lewis Tunstall, Leandro von Werra, Thomas Wolf "Natural Language Processing with Transformers", O'Reilly Media, 2022
2. Patterson, David et al "The Carbon Footprint of Large Neural Networks", Google Research Paper, 2021.

Reference Books

1. Andrej Karpathy "A Recipe for Training Neural Networks
2. Hugging Face Documentation – PEFT and Transformers

Useful Links

- 1 <https://arxiv.org/abs/2104.10350>
- 2 <https://huggingface.co/docs/peft>
- 3 <https://huggingface.co/docs/peft>

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6M006	MDM-VI (Multimodelling Development Systems)	2	0	0	2

Prerequisites for the course

1.	Basic knowledge of dynamical systems.
2.	Learn different modelling paradigms and their integration.

Prior Reading Material/useful links

1	https://www.anylogic.com/resources/documentation/
2	https://cloud.google.com/use-cases/multimodal-ai

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able to Understand the principles and benefits of multimodelling development systems.
2	CO2	Students will be able to Apply different modelling paradigms such as continuous, discrete, and hybrid models.
3	CO3	Students will be able to Use multimodel development tools (Modelica, Simulink, AnyLogic) for simulation.
4	CO4	Students will be able to Perform optimization, calibration, and sensitivity analysis in multimodel systems.
5	CO5	Students will be able to Design and implement hybrid and real-time multimodel applications in engineering and AI contexts.

Syllabus:

Course Contents		Hours
Unit I	Introduction to Multimodelling: Overview of multimodelling and model-driven engineering, Types of models: continuous, discrete, stochastic, hybrid, Benefits and challenges of multimodelling, Applications in engineering, AI, and simulation systems.	6

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Unit II	Modelling Paradigms and Techniques: Continuous-time and discrete-event modeling, System dynamics and agent-based modeling, Co-simulation concepts and frameworks, Model validation, verification, and consistency checking.	6
Unit III	Multimodel Development Tools: Introduction to multimodelling software and platforms (e.g., Modelica, Simulink, AnyLogic), Modelling libraries and component reuse, Integration of heterogeneous models, Simulation management and monitoring.	6
Unit IV	Optimization and Simulation: Parameter estimation and calibration techniques, Optimization in multimodel systems, Multi-objective modelling and analysis, Scenario-based simulations and sensitivity analysis.	6
Unit V	Advanced Topics & Applications: Hybrid models combining AI/ML with physical simulations, Real-time multimodel systems, Digital twins and cyber-physical systems, Case studies in engineering, healthcare, and transportation	6

Text Books	
1.	Peter Fritzson "Principles of Object-Oriented Modeling and Simulation with Modelica, 2nd Edition, Wiley, 2014
2.	AnyLogic Official Documentation Multimodel Simulation Guide
Reference Books	
1.	Grieves, Michael "Digital Twin: Manufacturing Excellence through Virtual Factory Replication", 2019
2.	Zeigler, Bernard P. et al. "Theory of Modeling and Simulation", 3rd Edition, Academic Press, 2000
Useful Links	
1	https://www.anylogic.com/resources/documentation
2	https://www.leewayhertz.com/multimodal-model/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6L001	Cloud Computing Lab	0	0	2	1

Prerequisites for the course

1.	Understand Basics of Discrete Mathematics
2.	Fundamental of SQL.

Prior Reading Material/useful links

1.	https://shanmugha.edu.in/pdf/dept/cse/LAB_MANUALS/Cloudcomputinglabmanual_cse.pdf
2.	https://www.stannescet.ac.in/cms/staff/qbank/CSE/Lab_Manual/CS8711_CLOUD%20COMPUTING%20LABORATORY-850713797_CC%20LAB%20MANUAL%20(2).pdf

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will be able to Configure various virtualization tools such as Virtual Box, VMware workstation.
2	CO2	Student will be able to Design and deploy a web application in a PaaS environment.
3	CO3	Student will be able to Learn how to simulate a cloud environment to implement new schedulers.
4	CO4	Student will be able to Install and use a generic cloud environment that can be used as a private cloud.
5	CO5	Student will be able to Manipulate large data sets in a parallel environment..

Course Contents List of Practical's

		Hours
0	Introduction to Cloud Computing Lab.	2
1	Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8.	2
2	Install a C compiler in the virtual machine created using virtual box and execute Simple Programs..	2

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3	Install Google App Engine. Create hello world app and other simple web applications using python/java.	2
4	Register with Amazon AWS and create a windows/linux instance and connect with RDP and create S3 buckets	2
5	Use GAE launcher to launch the web applications..	2
6	Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim	2
7	Find a procedure to transfer the files from one virtual machine to another virtual machine.	2
8	Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version.	2
9	Installation and Configuration of Just cloud.	2
10	Install Hadoop single node cluster and run simple applications like wordcount.	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books

1.	Cloud Computing Thomas Erl, Zaigham Mahmood, Ricardo Puttini Pearson
2.	Cloud Computing: Concepts, Technology, Security & Architecture, 2nd Edition - Pearson

Reference Books

1.	Cloud Computing Computerized Notes for Computer Science & Engineering by Riyaz .
2.	Cloud Computing Networking: Theory, Practice, and Development by Lee Chao · 2015

Useful Links

1.	https://srmvalliammai.ac.in/wp-content/uploads/2022/05/it8711-foss-and-cloud-computing-lab-manual.pdf
2.	https://files.mlrit.ac.in/curriculum/mtech/CSE-MLR18/1-1/CLOUD-COMPUTING-LAB.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS6L002	NoSQL Databases Lab	0	0	2	1

Prerequisites for the course

1.	Basics Knowledge of Computer programming.
2.	Introduction to Computer programming Lab.

Prior Reading Material/useful links

1.	https://www.javatpoint.com/machine-learning
2.	https://www.tutorialspoint.com/machine_learning/index.htm
3.	https://onlinecourses.nptel.ac.in/noc23_cs18/preview

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will be able to choose the concepts of computational intelligence in Machine Learning for Designing.
2	CO2	Students will be able to classify the data based on Bayes Theorem and other techniques.
3	CO3	Students will be able to apply ANN Techniques to address the real time problems.
4	CO4	Students will be able to build new software for Real time problems using Reinforcement Learning.
5	CO5	Students will be able to design different Board positions and different applications by using Analytical Learning.

List of Practical's

	Course Contents	Hours
0	Introduction of NoSQL Databases Lab.	2
1	Installation of NoSQL Databases	2
2	Demonstrate various data pre-processing techniques for a given dataset	2
3	Implement Dimensionality reduction using Principle Component Analysis (PCA) method.	2
4	Write a Python program to demonstrate various Data Visualization	2

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	Techniques.	
5	Implement Simple and Multiple Linear Regression Models.	2
6	Develop Logistic Regression Model for a given dataset.	2
7	Develop Decision Tree Classification model for a given dataset and use it to classify a new sample.	2
8	Implement Naïve Bayes Classification in Python	2
9	Build KNN Classification model for a given dataset.	2
10	Write a python program to implement K-Means clustering Algorithm.	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books

1.	Tom M. Mitchell, "Machine Learning", McGraw-Hill, 2010
2.	Bishop, Christopher. Neural Networks for Pattern Recognition. New York, NY: Oxford University Press, 1995.
3.	Introduction to Machine Learning, Fourth Edition By Ethem Alpaydin-2020-MIT Press

Reference Books

1.	Ethem Alpaydin, (2004) "Introduction to Machine Learning (Adaptive Computation and Machine Learning)", The MIT Press.
2.	T. astie, R. Tibshirani, J. H. Friedman, "The Elements of Statistical Learning", Springer(2nd ed.), 2009

Useful Links

1.	https://www.jnit.org/wp-content/uploads/2020/04/Machine-Learning-Lab-Manual.pdf
2.	https://deepakdvallur.weebly.com/machine-learning-laboratory.html
3.	https://mrcet.com/pdf/Lab%20Manuals/MACHINE%20LEARNING%20LAB.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6L003	Full Stack Development	0	0	4	2

Prerequisites for the Course

1.	Fundamental of Web Designing.
2.	Basic concepts of Python Programming.

Prior Reading Material/Useful Links

1.	https://www.w3schools.com/whatis/whatis_fullstack.asp
2.	https://www.coursera.org/articles/full-stack-developer
3.	https://www.geeksforgeeks.org/what-is-full-stack-development/

Course Outcomes

Sr. No	Course Outcome No.	CO statement
1	CO1	Students will develop an understanding of the formalistic (aesthetic) aspects of design and visual communication
2	CO2	Students will demonstrate cross-platform (web, mobile, broadcast, print) storytelling skills.
3	CO3	Students will become familiar with graphic design and/or game theory and be able to apply this theory to real world projects.
4	CO4	Students will develop and understanding of information design and usability as it applies to interactive media projects.
5	CO5	Students will utilize coding and software tools to analyze and present data in a professional manner that could be translated to web-based or app-based media.

Theory

Course Contents		Hours
Unit I	Front-End Development Basics: Introduction to Web Development: Client vs Server, HTML5: Structure, Forms, Semantic Elements, CSS3: Styling, Flexbox, Grid, Media Queries, JavaScript Basics: Variables, Functions, Events, DOM Manipulation, Introduction to Front-End Frameworks (e.g., Bootstrap or Tailwind CSS), Responsive Web Design	5

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	Principles	
Unit II	Back-End Development Essentials: Introduction to Server-Side Development, Node.js and Express.js Basics, HTTP Methods and RESTful APIs, Middleware, Routing, and API Endpoints, Introduction to databases: SQL vs NoSQL, CRUD operations with MongoDB or MySQL	5
Unit III	Full Stack Integration: Connecting Front-End to Back-End, Asynchronous Programming & Fetch/AJAX, Authentication Basics (Sessions, JWT), Project Structure & Environment Setup, Deploying a Full Stack App (using platforms like Heroku, Vercel, or Netlify), Mini Project: Build and Deploy a Full Stack Web Application	5

Activity

Sr. No.	Activity Topic
1	create a website using HTML CSS and JavaScript.
2	build a Chat module using HTML CSS and JavaScript.
3	create a simple calculator Application using React JS.
4	create a voting application using React JS.
5	create and Build a Password Strength Check using JQuery.
6	create and Build a star rating system using JQuery.
7	Create a Simple Login form using React JS.
8	Using the CMS users must be able to design a web page using the drag and drop method.
9	Create a project on Grocery delivery application.
10	Connecting our TODO React js Project with Firebase.

Text Books

1.	Web Development for beginners: Learn HTML/CSS/Javascript step by step with this Coding uide, Programming Guide for beginners, Website development, White Belt
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	Mastery, ISBN9781667003771.
2.	The Road to React: Your journey to master React.js in JavaScript (2021 Edition), Kindle Edition.
3.	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5, Publisher Shroff Publishers & Distributers, ISBN-13 978-9352130153
4.	Full Stack Web Development For Beginners Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP

Reference Books

1.	Mastering Html, Css&Javascript Web Publishing , BPB Publications , ISBN-13 978-8183335157
2.	A Complete Overview On: Web-development, Notion Press, ISBN-13978-1685098407.
3	Full Stack Development with JHipster Build Full Stack Applications and Microservices with Spring Boot and Modern JavaScript Frameworks, 2nd Edition By <u>Deepu K Sasidharan</u> , <u>Sendil Kumar N</u> · 2020
4	Full Stack JavaScript Development With MEAN MongoDB, Express, AngularJS, and Node.JS By <u>Colin J Ihrig</u> , <u>Adam Bretz</u> · 2014

Useful Links

1.	https://www.google.co.in/books/edition/Hands_On_Full_Stack_Development_with_Go/HHWPDwAAQBAJ?hl=en&gbpv=1&dq=Full+Stack+Development+book&printsec=frontcover
2.	https://www.google.co.in/books/edition/Modern_Full_Stack_Development/XLfZDwAAQBAJ?hl=en&gbpv=1&dq=Full+Stack+Development+book&printsec=frontcover
3.	https://www.youtube.com/watch?v=nu_pCVPKzTk
4.	https://careerfoundry.com/en/blog/web-development/what-is-a-full-stack-web-developer/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6P002	Mini Project-II	0	0	2	1

Prerequisites for the course

1.	Understand the creation of new technologies or innovations.
2	Learning business opportunities.

Prior Reading Material/useful links

1.	https://www.youtube.com/watch?v=p8e5ZEPRmx0
2.	https://www.youtube.com/watch?v=if_z7pMA85g
3.	https://www.youtube.com/watch?v=V5yv5TNpiLE

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will able to creation of new technologies or innovations.
2	CO2	Students will able to address complex problems or challenges
3	CO3	Students will able to develop new computational models or simulations that help advance their research.
4	CO4	Students will able to create new products or services that can be commercialized.
5	CO5	Students will able to impact various aspects of society, from improving productivity and efficiency to enhancing user experience and addressing social challenges.

This is continuous work to the Mini Project phase I. Every students will have to submit a completed report (3 copies)* of the project work. Report preparation guidelines should be followed as per given format. The students will prepare a power point presentation of the work. Panel of examiners comprising of guide, internal examiner, senior faculty, external examiner, etc. will assess the performance of the students considering their quality of work.

Mini Project Phase II

1. Coding/Implementation.
2. Use cases.
3. Testing/Trouble shooting.
4. Data dictionary/ Documentation.

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5. Finalization of project in all respect.

6. *(For guide, Personal copy, Departmental library.)

7. In a presentation, the students should focus to clarify problem definition and analysis of the problem.

Text Books

1.	Planning and Implementing Your Final Year Project — with Success! By <u>Mikael Berndtsson</u> , <u>Jörgen Hansson</u> , <u>B. Olsson</u> , <u>Björn Lundell</u> · 2013
2.	Computer Science Project Work Principles and Pragmatics By <u>Sally Fincher</u> , <u>Marian Petre</u> , <u>Martyn Clark</u> · 2001
3.	Thesis Projects, A Guide for Students in Computer Science and Information Systems By <u>Mikael Berndtsson</u> · 2008

Reference Books

1.	SOFTWARE ENGINEERING PROJECT MANAGEMENT By <u>Bill Brykczynski</u> , <u>Richard D. Stutz</u> · 2006
2.	Software Engineering Body of Knowledge By <u>IEEE Computer Society</u> · 2014
3.	Quality Software Project Management By <u>Robert T. Futrell</u> · 2002

Useful Links

1.	https://www.google.co.in/books/edition/Project_Summaries/0ggTG_ya8acC?hl=en&gbpv=1&dq=major+project+objective+in+computer+science+engineering&pg=PA78&printsec=frontcover
2.	https://www.google.co.in/books/edition/Real_World_Software_Projects_for_Compute/X6seEAAQBAJ?hl=en&gbpv=1&dq=major+project+objective+in+computer+science+engineering&printsec=frontcover
3.	https://www.logicraysacademy.com/blog/final-year-projects-for-cse/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS5J004	JDCAP IV- Skills for Recruitment Success	2	0	0	Audit

Prerequisites for the course

1.	Ability to observe patterns and relationships in data or figures.
2.	Basic understanding of English grammar and vocabulary.
3.	Interest in solving puzzles or logical problems.

Prior Reading Material/useful links

1	https://www.learntheta.com/aptitude-preparation
2	https://pdf.exampundit.in/quantitative-aptitude
3	https://www.placementpreparation.io/blog/best-websites-to-learn-quantitative-aptitude/

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able to Apply logical reasoning techniques to solve verbal and non-verbal reasoning questions.
2	CO2	Students will be able to Analyse complex reasoning problems to identify patterns, sequences, and logical connections.
3	CO3	Students will be able to Evaluate statements, arguments, and conclusions to arrive at valid logical decisions.
4	CO4	Students will be able to Inculcate cross cultural verbal communication and to develop strategic thinking , global communication, presentation and interview readiness.
5	CO5	Students will be able to Develop boardroom interview techniques, fellowship panels and research defense as well as students should be able to handle multi-round interview challenges.

Syllabus:

Course Contents		Hours
Unit I	Series and sequences, Analogy, Inequality, Mirror Image, Syllogism and Logical deduction: Number series, Alphabetical series, Mixed series, Missing and wrong term series. Word analogy, Number analogy and Letter analogy. Venn Diagram Approach, Syllogism (All, Some, None, Either–Or	6



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KATOL ROAD, NAGPUR

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Department of Computer Science & Engineering

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	Cases), Possibility & Either–Or Cases, Statements and Conclusions, Statements and Assumptions, Statements and Arguments	
Unit II	Coding -Decoding, Distance & Direction sense, Blood Relation & Ranking, Seating arrangement & Puzzle: Letter coding, number coding, Symbol coding and substitution coding. Direction Test – Distance, Angle, Rotation. Shadow and Mirror-based Direction Questions, Family Tree Representation, Puzzles on Blood Relations, Ranking Arrangement (Vertical, Horizontal), Comparison Type Questions, Linear Arrangement (Single, Double Row), Circular Arrangement (Facing Inward/Outward), Square/Rectangular Arrangement, Floor Puzzle, Age Puzzle, Scheduling Puzzle.	6
Unit III	Part-A Advanced Verbal Ability: Corporate vocabulary, negotiation language, and persuasion skills. Critical reading, inference-based comprehension, and abstract reasoning. Part-B Analytical & Critical Essay Writing ➤ Data-driven essays, policy analysis, and industry white papers. ➤ Persuasive writing with counterarguments and rebuttals.	6
Unit IV	Part-A- Strategic Group Discussion: ➤ GD leadership roles, handling disruptions, fact-based interventions. ➤ Case-based GD and decision-making simulations. Part-B High-Stakes Personal Interview ➤ Industry-specific technical + HR interview integration. ➤ Negotiating job offers & salary discussions. ➤ Cross-cultural interview etiquettes for global roles.	6

Text Books

- | | |
|----|--|
| 1. | R.S. Aggarwal – A Modern Approach to Logical Reasoning |
| 2. | Quantitative Aptitude and Reasoning By R.V. Pravin. |

Reference Books

- | | |
|----|--|
| 1. | A textbook of communication skill (2023), Author- Miss Vachche |
| 2. | Arun Sharma – Logical Reasoning for CAT |

Useful Links

- | | |
|---|---|
| 1 | https://prepinsta.com/ |
| 2 | https://www.indiabix.com/ |

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JDcoem, Nagpur

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CSE Dept.
JDcoem, Nagpur

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Activity & Subject Allotment 2025-26

Sr. No	Category of Subject	Course Code	Course Name	Activity	Subject Incharge
1	PCC	CS6T001	Cloud Computing	Technical Tool	Prof. Ragini Khobragade
2	PCC	CS6T002	Software Testing & Quality Assurance	Technology Exploration	Dr. Amit Welekar
3	PCC	CS6T003	AI & Applications	Internship	Dr. Ashutosh Lanjewar
4	PEC-III	CS6E003A	PEC-III (Middleware Technology)	Animated PPT/Video	Prof. Ragini Khobragade
5	PEC-III	CS6E003B	PEC-III (Soft Computing)	Animated PPT/Video	Prof. Sweta Kahurke
6	PEC-III	CS6E003C	PEC-III (Digital Image Processing)	Animated PPT/Video	Prof. Pooja Sarangpure
7	PEC-III	CS6E003D	PEC-III (Deep Learning)	Animated PPT/Video	Prof. Gajanan Wankhade
8	PEC-IV	CS6E004A	PEC-IV (No SQL New SQL)	Open Book Test	Prof. Mittal Patne
9	PEC-IV	CS6E004D	PEC-IV (Robotics Automation System)	Open Book Test	Prof. Gajanan Wankhade
10	PEC-V	CS6E005A	PEC-V (Data Warehousing & Mining)	Video/ Reel/ Blog Writing	Prof. Achal Wani
11	PEC-V	CS6E005C	PEC-V (Augmented Reality)	Video/ Reel/ Blog Writing	Prof. Nitin Raut
12	MDM-V	CS6M005	LLM Fine Tuning & Optimization	Technology Enhancement	Prof. Karishma Bobade
13	MDM-VI	CS6M006	Multimodelling Development Systems	Simulation & Modelling	Prof. Sweta Kahurke
14	PCC	CS6L001	Cloud Computing Lab	-	Ms. Gayetri Gawande/ Mr. Om Butle
15	PCC	CS6L002	NoSQL Databases Lab	-	Prof. Pooja Sarangpure
16	VSEC	CS6L003	Full Stack Development	-	Ms. Gayetri Gawande/ Mr. Om Butle
17	ELC	CS6P003	Field Project / Mini Project-III	As per Rubrics	Prof. Anuja Ghasad
13	JDCAP	CS6T002	JDCAP IV: Skills for Recruitment Success	-	TNP



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